Automated Skin Disease Detection from Dermoscopic Images with Explainability Analysis

Research Proposal

Submitted by:

Tashmika Pillay

Supervisor:

Prof Serestina Viriri

University of KwaZulu-Natal
School of Mathematics, Statistics and Computer Science

Bachelor of Science Honours in Computer Science

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Abstract

Skin diseases are part of a major public health concern, with melanoma and other pigmented lesions carrying a serious mortality risk if not detected early. Dermoscopy is the clinical gold standard for examining skin lesions, but requires specialist expertise that is often unavailable in many healthcare settings, especially in resource-limited environments such as the rural areas of South Africa. This research project aims to design and evaluate a novel lightweight hybrid deep learning model for automated multi-class classification of dermoscopic images, with an integrated explainability analysis to ensure clinical interpretability.

The proposed approach revolves around a lightweight hybrid Convolutional Neural Network (CNN)–Transformer architecture integrating the local feature extraction characteristics of CNNs with the ability of Transformer Encoders to model global context. The HAM10000 dermoscopic benchmarked dataset [1], publicly available and consisting of 10,015 images across seven diagnostic categories, will be used to train and evaluate the model. The proposed work extends a CNN–Transformer architecture, with the main contributions being the lightweight redesign, systematic multi-method explainability analysis, and an exploratory few-shot and zero-shot learning component for under-represented disease categories. Explainability analysis will then be conducted using Grad-CAM [2], SHAP [3] and native attention map visualization. This will validate if model predictions are driven by clinically significant features aligned with standard dermoscopic diagnostic criteria.

This research is an open-ended system design problem rather than a simple comparative benchmark, with a hybrid architecture serving as the primary methodological contribution. A web application that inputs dermoscopic images and outputs classification results with Grad-CAM heatmap overlays will serve as the software demonstration deliverable.

Keywords: Detection of Skin Diseases; Dermoscopic Images; CNN-Transformer; Explainable Analysis; Lightweight Model.

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1 Introduction

Skin diseases affect hundreds of millions of people across all age groups and geographical contexts [4]. Melanoma, specifically, carries a significant mortality risk, with survival outcomes depending on the diagnosis stage. While advanced melanoma is associated with substantially lower survival rates, early-stage disease is highly treatable, [5].

The burden of skin cancer is severe in regions with high ultraviolet exposure, such as Australia [6]. The HAM10000 dataset used in this study is derived from populations in Austria and Australia, reflecting clinical data from regions with a high prevalence of skin cancer [1].

Automated deep learning systems are capable of classifying dermoscopic images into clinically meaningful disease categories. This creates a path towards scalable and cost-effective diagnostic support. CNNs have historically dominated automated dermoscopic classification, reaching dermatologist-level accuracy on several benchmarks [5]. More recently, Vision Transformers (ViTs) have achieved competitive results on medical imaging tasks by using self-attention to model long-range dependencies between image patches [7]. Hybrid architectures pairing convolutional feature detection with the broad contextual modelling of Transformers offer a promising direction for medical image analysis [8], yet remain under-explored in the dermoscopy domain.

Despite substantial research activity, critical gaps exist. Most implemented models operate as black boxes, providing classification outputs without explanation. This is a fundamental barrier to clinical trust and regulatory approval [9]. Moreover, the relationship between architectural design and the clinical interpretability of model explanations has not been examined extensively. This project proposes to address these gaps through the design of a novel lightweight hybrid CNN–Transformer architecture, building upon an existing open-source baseline, combined with a systematic explainability analysis to ensure that model decisions are grounded in clinically meaningful visual features.

The remainder of this proposal is structured as follows. Section 2 states the motivation of this research. Section 3 states the premises of research. Section 4 explores related work. Section 5 defines the problem statement. Section 6 presents the research questions. Section 7 states the aims and objectives. Section 8 describes the proposed methodology. Section 9 presents the proposed system architecture. Section 10 outlines expected results and deliverables. Section 11 addresses ethics considerations. Section 12 presents the project timetable and Section 13 the estimated budget.

2 Motivation

The motivation for this research is both clinical and technical. From a clinical perspective, dermoscopic diagnosis creates a significant access gap that affects patients in disadvantaged communities — a challenge of direct relevance in the South African healthcare system where specialist access is limited [10]. An automated, explainable classification system could potentially serve as a decision-support tool for general practitioners and community health workers, enabling earlier referral and treatment.

From a technical perspective, the field currently lacks a comprehensive investigation into how lightweight

hybrid architectures perform on multi-class dermoscopic classification under class imbalance. Moreover, a model may achieve high overall accuracy while relying on clinically irrelevant image regions, which is unacceptable for medical deployment [9]. These gaps motivate the dual focus of this project on efficient architecture design and systematic explainability analysis.

On a personal level, the integration of computer vision, medical imaging, and explainable AI represents an area of strong research interest. This domain inclines towards the application of computational methods within medical and clinical areas, particularly in supporting diagnostic decision-making and improving accuracy. The potential for this work to produce tangible clinical impact in healthcare environments similar to those encountered in South Africa provides additional motivation to pursue this research direction.

3 Premises of the Research

Research is grounded in established scientific, clinical, and methodological principles to ensure the reliability and credibility of the results. This research follows a systematic approach, drawing on validated datasets, recognised clinical frameworks, and reproducible experimental design:

- **Clinical validity of the HAM10000 dataset:** The HAM10000 dataset was curated by the International Skin Imaging Collaboration (ISIC) and validated by board-certified dermatologists [1]. It is the most widely used publicly available benchmark in automated dermoscopic classification and provides a clinically grounded evaluation basis.
- **Established diagnostic criteria:** The ABCDE rule (Asymmetry, Border, Colour, Diameter, Evolution) and the seven-point checklist form the internationally validated diagnostic frameworks used by dermatologists worldwide. These criteria provide the clinical standard against which the explainability analysis will be assessed.

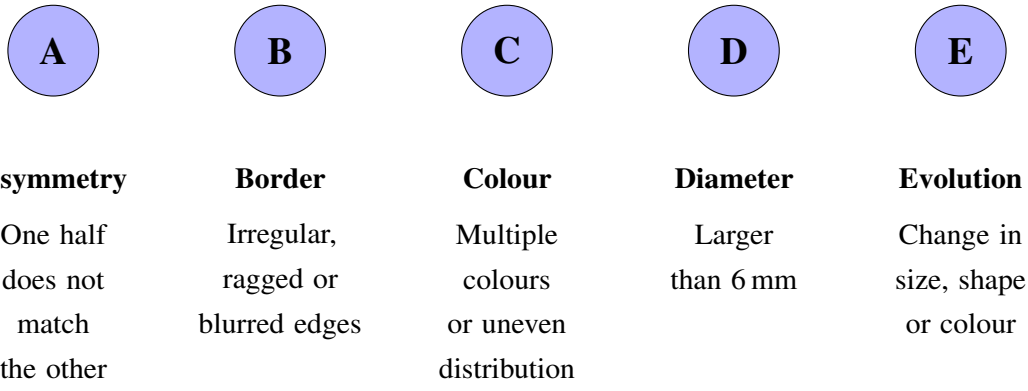


Figure 1: The ABCDE rule serves as the clinical reference standard against which the explainability outputs of the proposed model will be assessed.

- **Transfer learning from large-scale pretraining:** The use of ImageNet-pretrained weights as initialization has been demonstrated to improve performance on small medical imaging datasets by transferring rich visual representations [11].
- **Open-source architectural baseline:** Building upon an existing, publicly available and validated

open-source implementation reduces the implementation risks and allows for the novel contributions - lightweight redesign, systematic explainability, and few/zero-shot learning- to be clearly distinguished from prior work.

- **Reproducibility through controlled experimentation:** All architectural evaluations will be conducted under identical experimental conditions — same dataset, preprocessing, training protocol, and evaluation metrics.

4 Related Work

4.1 Esteva et al. (2017) — Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks

Esteva et al. [5] trained a single deep CNN on 129,450 clinical images spanning 757 disease classes, then tested the system against 21 board-certified dermatologists on two binary classification tasks: keratinocyte carcinoma versus benign seborrheic keratosis, and malignant melanoma versus benign nevi. The study aimed to determine whether an automated deep learning system could match specialist-level diagnostic accuracy. The authors employed transfer learning from an ImageNet-pretrained Inception v3 architecture, fine-tuned end-to-end on their dataset using a novel disease taxonomy and partitioning algorithm. No hand-crafted features were required; the model was trained directly on raw image pixels and disease labels. The partitioning algorithm mapped individual diseases into 757 fine-grained training classes, and performance was evaluated using sensitivity–specificity ROC curve analysis benchmarked against dermatologist responses. The CNN achieved an AUC exceeding 91% across all three tasks, matching or exceeding the performance of all 21 tested dermatologists, leading the authors to conclude that deep CNNs trained at sufficient scale can replicate specialist-level diagnostic accuracy and could extend dermatological care through mobile devices.

This work established the clinical viability of CNN-based skin disease classification and motivates this project directly. However, three noteworthy differences exist. First, Esteva et al. used a proprietary dataset unavailable to the research community; this project uses the publicly available HAM10000 dataset to ensure reproducibility. Second, their study addressed binary classification tasks, whereas this project performs seven-class multi-label classification. Third, no explainability analysis was conducted; this project addresses that gap through Grad-CAM, SHAP, and attention map visualization.

4.2 Jaycee-404 (2023–2024) — Hybrid EfficientNetB0 + ViT on HAM10000

A hybrid CNN–Transformer proof-of-concept combining EfficientNetB0 and a Vision Transformer encoder for seven-class classification on the HAM10000 dataset was proposed by Jaycee-404 [12]. The implementation uses EfficientNetB0 as a CNN feature extractor whose output is tokenised and passed into a compact Transformer encoder block, with standard cross-entropy loss and no explicit class imbalance handling. After iterative refinement, the pipeline achieves a reported validation accuracy of 86.82%, demonstrating the viability of the hybrid paradigm on this benchmark, though performance lags behind stronger baselines in the literature due to the absence of imbalance handling and limited

architectural optimisation.

The hybrid CNN–Transformer paradigm demonstrated in this work confirms the viability of combining convolutional feature extraction with Transformer-based attention for dermatological classification, motivating the architectural direction of this research. This study advances beyond that foundation through: (1) a full architectural redesign targeting lightweight deployment via a MobileNet-style backbone and compressed Transformer encoder; (2) a multi-strategy class imbalance handling pipeline; (3) systematic multi-method explainability analysis assessed against clinical ABCDE criteria; (4) an ablation study quantifying individual design contributions; and (5) exploratory few-shot and zero-shot learning for under-represented lesion categories. Together, these contributions constitute a substantive methodological extension of the hybrid paradigm rather than an incremental modification.

4.3 Tschandl, Rosendahl and Kittler (2018) — The HAM10000 Dataset

Tschandl et al. [1] curated and publicly released the HAM10000 dataset — a publicly available, multi-source dermoscopic image collection covering seven diagnostic categories of pigmented skin lesions, collected over 20 years from clinical sites in Vienna, Austria and Queensland, Australia. Images were gathered from multiple sites using different dermoscopic devices, with an automated image-type filtering pipeline using a fine-tuned InceptionV3 classifier separating dermoscopic images from clinical photographs and overviews, followed by manual quality review. Ground truth was established through four methods: histopathological biopsy (for over 50% of cases), in-vivo confocal microscopy, digital dermatoscopic follow-up, and expert consensus for typical benign lesions. The final dataset covers: melanocytic nevi (nv, 6,705 images), melanoma (mel, 1,113), benign keratosis (bkl, 1,099), basal cell carcinoma (bcc, 514), actinic keratosis (akiec, 327), vascular lesions (vasc, 142), and dermatofibroma (df, 115). HAM10000 subsequently became the standard benchmark for multi-class dermoscopic classification research and was adopted as the primary dataset for the ISIC 2018 challenge.

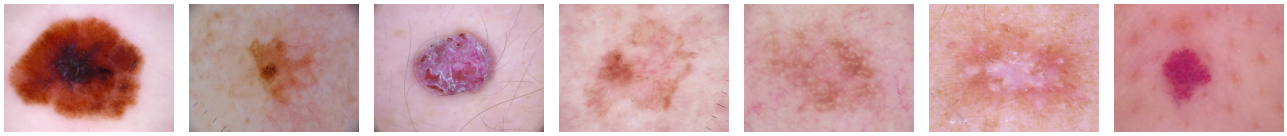
HAM10000 is the primary dataset used in this project, making this paper a foundational reference. The published class distribution (6,705 nevi versus 115 dermatofibromas) reveals a severe class imbalance that directly motivates the multi-strategy imbalance handling approach in this project's methodology. The dataset's known demographic bias towards lighter skin tones from European and Australian populations will be explicitly acknowledged as a limitation in all outputs.

4.4 Adebiyi et al. (2024) — Accurate Skin Lesion Classification Using Multimodal Learning on HAM10000 and ISIC 2017

Adebiyi et al. [13] investigated whether combining dermoscopic images with structured patient metadata — particularly age, sex, and lesion localisation — through a multimodal deep learning model could outperform image-only CNN models for skin lesion classification on HAM10000 and ISIC 2017. The authors compared four image-only CNN baselines (InceptionV3, ResNet50, DenseNet121, EfficientNetB0) against the ALBEF multimodal model, which fuses a Vision Transformer image encoder (ViT-B/16) with a BERT text encoder through an attention-based alignment mechanism to jointly process lesion images and clinical text. All models were trained with identical hyperparameters including batch

size, Adam optimizer, number of epochs, and learning rate, with data augmentation including random flips, rotation, and colour jitter, using a 70/20/10 train–validation–test split. An ablation study assessed the contribution of each individual metadata feature to overall performance. The multimodal ALBEF model achieved the best results on HAM10000, with an accuracy of 0.9411 and AUROC of 0.9426, outperforming all image-only models, while EfficientNetB0 was the strongest image-only baseline at 0.9022 accuracy and 0.9191 AUROC.

This paper confirmed the strong performance of EfficientNet as a CNN baseline on HAM10000, supporting its use as a comparison architecture in this project, and demonstrates that combining image and clinical data improves classification. However, Adebiyi et al. did not propose a lightweight architecture suitable for resource-constrained deployment, and their explainability analysis — SHAP applied only to DenseNet121 — was limited and peripheral rather than a primary contribution. This project builds on the existing work by proposing a novel lightweight hybrid CNN–Transformer architecture and conducting a systematic, multi-method explainability analysis as a core research contribution.



Melanoma (mel)	Nevi (nv)	BCC (bcc)	AK (akiec)	Benign K. (bkl)	DF (df)	Vasc. (vasc)

Figure 2: Representative dermoscopic images from each of the seven diagnostic categories in the HAM10000 dataset. Images sourced from the HAM10000 dataset [1], distributed under CC-0 licence via the ISIC Archive.

4.5 Chen et al. (2021) — TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation

Chen et al. [8] proposed TransUNet, a hybrid architecture combining a CNN encoder with a Transformer-based bottleneck for medical image segmentation, evaluated on the Synapse multi-organ segmentation dataset and the ACDC cardiac dataset. The authors replaced the standard fully convolutional bottleneck in a U-Net architecture with a Transformer module, allowing the model to capture long-range dependencies between spatial positions, with CNN features tokenised and fed into a standard Transformer encoder before being decoded alongside skip connections. TransUNet outperformed both pure CNN and pure Transformer baselines across several segmentation benchmarks, demonstrating that combining the complementary strengths of both paradigms yields more effective results in medical image analysis than either approach independently.

TransUNet addressed segmentation rather than classification and targeted anatomical structures rather than skin lesions. This project applies the hybrid paradigm to multi-class dermoscopic classification, incorporates class imbalance handling strategies specific to dermoscopy, and includes a systematic explainability analysis component that was absent from TransUNet.

4.6 Lucieri et al. (2022) — ExAID: A Multimodal Explanation Framework for Computer-Aided Diagnosis of Skin Lesions

Lucieri et al. [9] developed ExAID, a framework for concept-based explainability in skin lesion classification that mapped model predictions to human-interpretable dermoscopic concepts derived from established clinical diagnostic criteria, enabling clinicians to understand model decisions in familiar diagnostic terms. The authors trained concept classifiers on annotated dermoscopic concepts and integrated them with the main classification network using concept activation vectors (CAVs), allowing explanations to be expressed in terms of which clinical concepts were active in a given prediction, evaluated on the PH2 dataset with expert concept annotations. ExAID produced clinically meaningful explanations that aligned with dermatologist reasoning, and user studies with clinicians indicated that concept-based explanations improved trust and diagnostic confidence compared to purely visual heatmap-based methods.

ExAID demonstrates the clinical value of explanations aligned with dermoscopic criteria, which directly motivates the explainability component of this project. However, ExAID required extensive concept annotation which limited its scalability. This project instead uses gradient-based (Grad-CAM [2]) and attribution-based (SHAP [3]) methods that do not require additional annotations, assessing alignment with the ABCDE rule post-hoc. The architectural contribution of a lightweight hybrid model is also absent from ExAID.

5 Problem Statement

Dermoscopic diagnosis of skin diseases faces an access gap that contributes to delayed detection and increased mortality in many healthcare settings globally, especially for conditions such as melanoma [10]. While deep learning systems have demonstrated clinical-level accuracy on skin disease classification benchmarks [5], two critical problems prevent widespread clinical adoption. First, existing approaches lack architectures that are simultaneously lightweight enough for resource-constrained deployment and accurate enough under the severe class imbalance inherent in dermoscopic datasets [1]. Second, the near-universal absence of interpretable model explanations prevents clinicians from integrating or trusting automated diagnoses [9]. This study therefore investigates how a lightweight hybrid CNN–Transformer architecture, developed from an existing open-source baseline, can be designed and validated for dermoscopic skin disease classification, and whether its predictions can be made clinically interpretable through systematic explainability analysis.

6 Research Questions

Main Research Question

How can a lightweight hybrid CNN-Transformer architecture be designed, implemented, and evaluated to achieve accurate and clinically interpretable multi-class classification of dermoscopic skin disease images from the HAM10000 dataset?

Sub-Research Questions

1. What preprocessing and class imbalance mitigation strategies are most effective for preparing the dataset for training deep learning classifiers?
2. How does the proposed lightweight hybrid CNN-Transformer architecture compare against EfficientNet-B3 and ViT-B/16 baselines, with regards to overall accuracy, F1-score per class, balanced accuracy and AUROC under matched experimental conditions?
3. Does the Grad-CAM, SHAP, and attention map visualizations produced by the hybrid model correspond to the clinically established dermoscopic diagnostic criteria defined by the ABCDE rule?
4. What are the independent contributions of class imbalance handling strategy, ImageNet pretraining, and input resolution to classification performance across architectures?
5. To what extent can few-shot and zero-shot learning strategies improve model generalisation when labelled training data for certain disease categories is limited?

7 Aim and Objectives

Aim

To design, implement and evaluate a lightweight hybrid CNN-Transformer deep learning architecture for accurate and clinically interpretable automated multi-class classification of dermoscopic skin disease images, with systematic explainability analysis to validate the clinical significance of model predictions.

Objectives

1. Conduct comprehensive preprocessing on the HAM10000 dataset and implement a multi-strategy class imbalance mitigation pipeline through class augmentation.
2. Model a lightweight hybrid CNN-Transformer architecture based on a compact MobileNet-style CNN to reduce the complexity of the Transformer encoder.
3. Implement Explainable analysis using Grad-CAM, SHAP, and native attention map visualization across all trained architectures and compare with the ABCDE rule diagnostic features.
4. Evaluate the proposed lightweight hybrid architecture through an ablation study by isolating and measuring the independent contributions.
5. Investigate the efficiency of Few-shot and Zero-shot learning in the classification of under-represented skin disease categories.

8 Research Methodology

This research follows an experimental deep learning research methodology comprising five sequential phases. The proposal is intentionally framed as flexible at this stage, allowing refinement as the research progresses.

8.1 Alignment Table

Table 1: Alignment of research questions, objectives, methodology, and deliverables

Research Question	**Objective**	**Methodology**	**Deliverable**
RQ1: Preprocessing and class imbalance	Objective 1	Hair removal; colour normalisation; patient-level data splitting; class-weighted loss; data augmentation; oversampling	Preprocessed dataset; documented preprocessing pipeline
RQ2: Hybrid model versus baselines	Objective 2	Extend lightweight hybrid CNN–Transformer architecture; EfficientNet-B3 and ViT-B/16 baselines; AdamW optimiser; matched training protocol; evaluation using accuracy, F1-score, and AUROC evaluation	Trained models; benchmark comparison table
RQ3: Explainability and clinical alignment	Objective 3	Grad-CAM; SHAP DeepExplainer; attention map extraction; qualitative ABCDE rule assessment	Heatmap visualizations; clinical alignment analysis
RQ4: Ablation study	Objective 4	Systematic ablation of class imbalance handling, ImageNet pretraining, and input image resolution	Design guidelines document
RQ5: Few-shot and zero-shot learning	Objective 5	Prototypical networks for few-shot; CLIP-based zero-shot; minority class experiments	Feasibility study report; preliminary experimental results

8.2 Phase 1: Dataset and Preprocessing

The HAM10000 dataset [1], accessed via the ISIC Archive, consists of 10,015 dermoscopic images across seven categories. The preprocessing includes: automated hair removal (Figure 3a demonstrates one of the input images) via morphological filtering and inpainting, colour space normalisation, resizing to 224×224 pixels; and patient-level train/validation/test splitting at a 70/15/15 ratio. As shown in Figure 3b, the dataset exhibits severe class imbalance. This will be addressed through a combination

of class-weighted cross-entropy loss, targeted augmentation of minority classes (random flips,rotation, colour jitter), and synthetic oversampling where appropriate.

Class-Weighted Cross-Entropy Loss

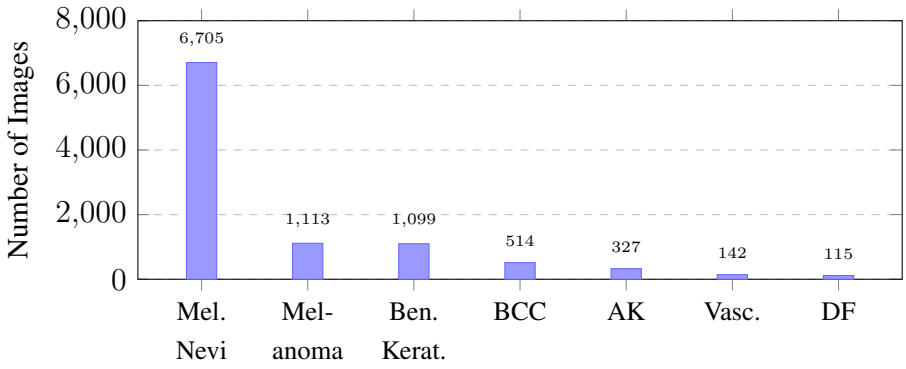
The class-weighted cross-entropy loss is defined as:

$$\mathcal{L} = -\sum_{c=1}^{C} w_c \sum_{i=1}^{N} y_{i,c} \log \hat{y}_{i,c} \tag{1}$$

where $w_c = \frac{N}{C \cdot N_c}$ is the weight for class c, N_c is the number of samples in class c, $y_{i,c}$ is the ground-truth label, and $\hat{y}_{i,c}$ is the predicted probability.



(a) Example dermoscopic image containing hair artefacts [1].



(b) Class distribution of the HAM10000 dataset. Melanocytic nevi (6,705) outnumber dermatofibroma (115) by approximately 58:1 [1].

Figure 3: Dataset characteristics: hair artefact example (left) and class distribution (right).

8.3 Phase 2: Architecture Design, Implementation and Evaluation

Using an EfficientNetB0 + ViT baseline, the primary contribution is a redesigned lightweight hybrid model incorporating a compact MobileNet-style CNN backbone for local feature extraction, patch tokenization of feature maps, and a reduced Transformer encoder with efficient attention. EfficientNet-B3 and ViTB/16 (both ImageNet-pretrained) will serve as comparison baselines. All models will be implemented in PyTorch under identical conditions. Performance will be measured using overall accuracy, per-class F1-score, balanced accuracy, macro-averaged AUROC and quadratic weighted kappa.

Evaluation Metrics

The following metrics are used to comprehensively evaluate model performance, particularly under class imbalance and multi-class diagnostic conditions. Given a confusion matrix with true positives (TP_c), false positives (FP_c), and false negatives (FN_c) for class c:

$$\text{Accuracy} = \frac{\sum_c TP_c}{N} \tag{2}$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3}$$

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{4}$$

$$\text{Macro F1} = \frac{1}{C} \sum_{c=1}^{C} F1_c \tag{5}$$

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c} \tag{6}$$

$$\text{AUROC} = \int_0^1 \text{TPR}(t)\, d\,\text{FPR}(t) \tag{7}$$

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}} \tag{8}$$

where N is the total number of samples, $C = 7$ is the number of diagnostic classes, $\text{TPR}(t)$ and $\text{FPR}(t)$ are the true and false positive rates at threshold t evaluated in a one-vs-rest scheme per class, O_{ij} is the observed frequency in the confusion matrix, E_{ij} is the expected frequency under chance agreement, and $w_{ij} = \frac{(i-j)^2}{(C-1)^2}$ are the quadratic weights penalising predictions further from the true class.

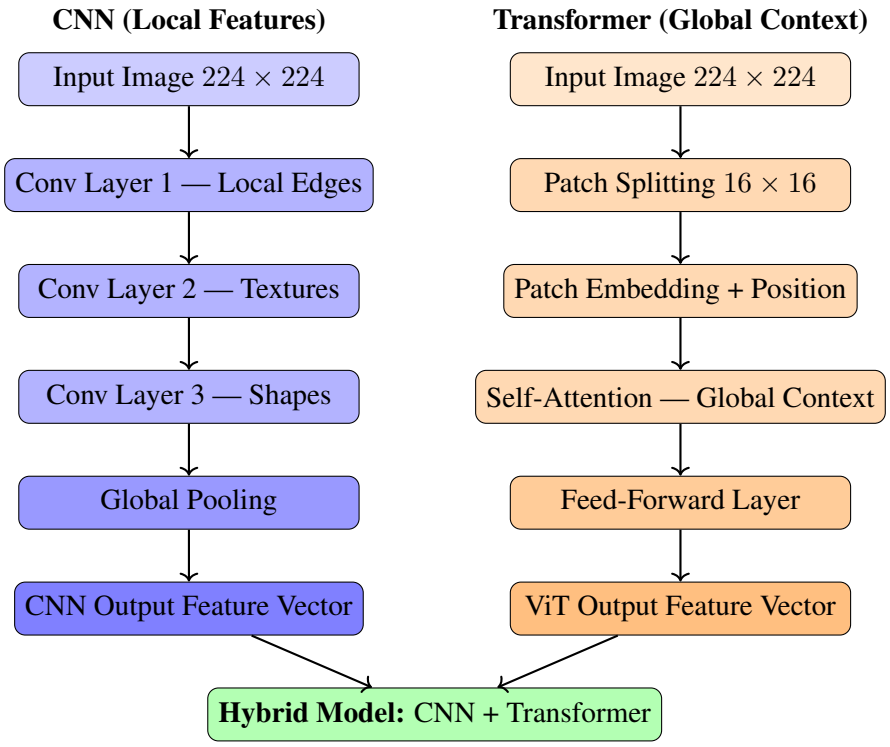


Figure 4: Comparison of CNN and Vision Transformer processing pipelines [7, 8].

8.4 Phase 3: Explainability Analysis

Three complementary XAI techniques will be applied across all architectures. Grad-CAM [2] generates spatial heatmaps higlighting regions that most influenced each classification decision. SHAP DeepExplainer [3] assigns per-pixel attribution scores quantifying each pixel's contribution toward or away from the predicted class. Native attention maps will be extracted from the Transformer components [7]. Clinical alignment will be evaluated by comparing highlighted regions against ABCDE rule criteria on representative samples from the seven disease categories.

Grad-CAM

The Grad-CAM heatmap for class c is computed as:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right), \quad \alpha^c_k = \frac{1}{Z}\sum_i \sum_j \frac{\partial S_c}{\partial A^k_{ij}} \tag{9}$$

where A^k is the activation map of the k-th convolutional feature, S_c is the class score before softmax, and Z is the number of spatial locations.

SHAP Values

SHAP values are grounded in the Shapley value from cooperative game theory. For a model f and input $\mathbf{x}$, the SHAP value for feature i is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \left[f_{S\cup\{i\}}(\mathbf{x}_{S\cup\{i\}}) - f_S(\mathbf{x}_S)\right] \tag{10}$$

where F is the set of all input features and S is a subset excluding feature i.

Self-Attention

The self-attention weight between query and key vectors is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{11}$$

where Q, K, V are the query, key, and value matrices and d_k is the key dimension used for scaling.

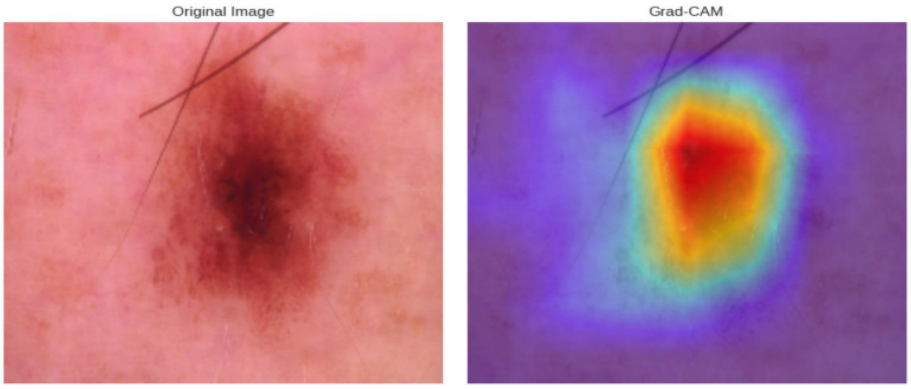


Figure 5: Illustrative example of Grad-CAM heatmap visualization on a dermoscopic image, showing the model's attention concentrated on the lesion region, with red areas indicating regions of highest influence on the classification decision. Figure reproduced from [14] for illustrative purposes.

8.5 Phase 4: Ablation Study

A systematic ablation study will evaluate the independent contributions of: class imbalance handling strategy; ImageNet pretraining (random initialisation vs. pretrained weights); and input resolution (112×112 vs. 224×224 vs. 448×448). Results will generate practical design guidelines.

8.6 Phase 5: Few-Shot and Zero-Shot Learning (Exploratory)

As an exploratory component, few-shot and zero-shot learning will be investigated for under-represented dermoscopic categories. Prototypical networks will be applied to minority classes (particularly dermatofibroma, 115 images) to assess whether per-class performance can be improved, while CLIP-based zero-shot classification will be explored using textual disease descriptions. This phase is subject to available time and computational resources.

9 Proposed System Architecture

Figure 6 represents the schematic overview of the proposed system pipeline. The system operates through seven sequential stages, followed by the software deliverable.

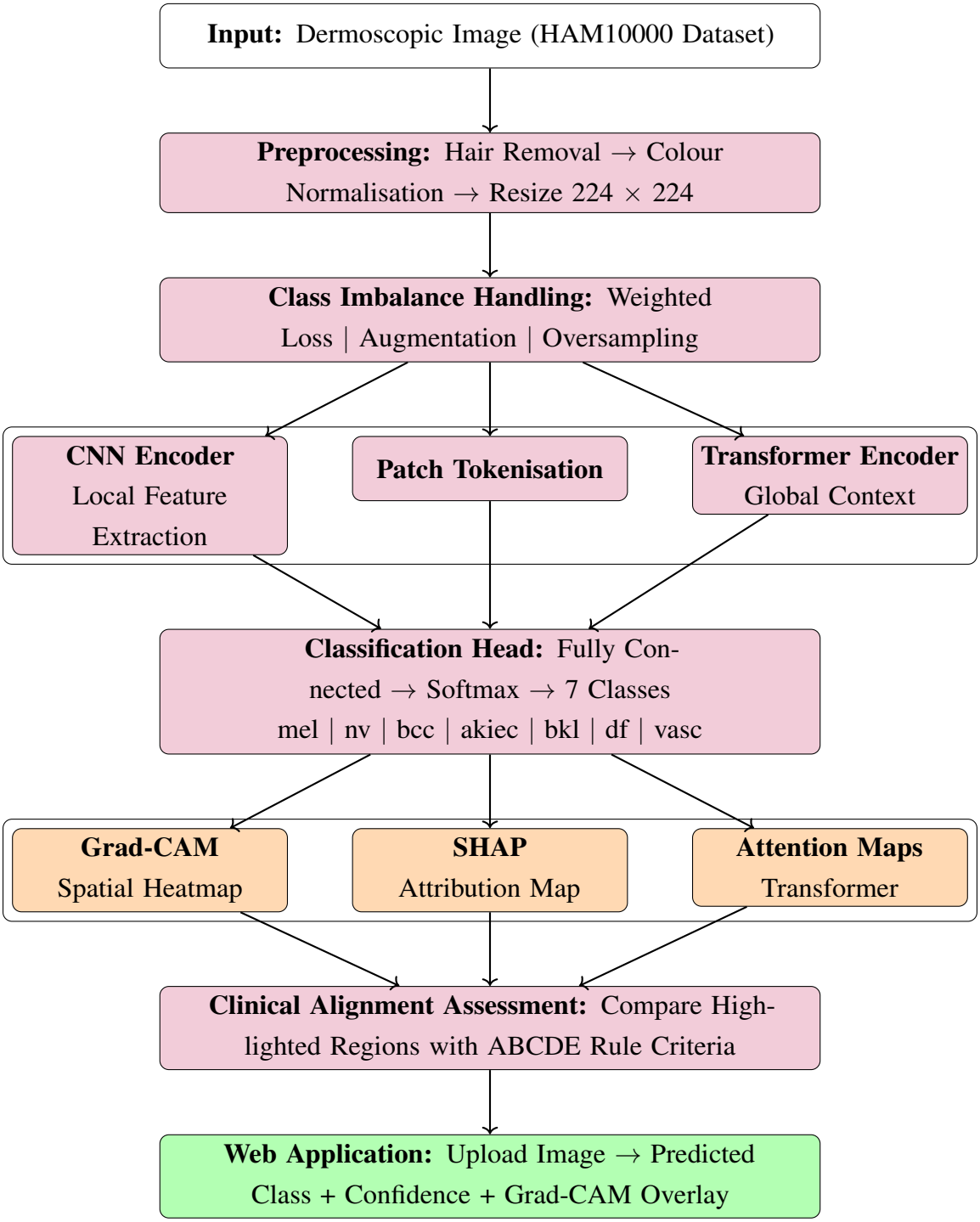


Figure 6: Proposed system architecture.

10 Expected Results (Deliverables)

Expected Results:

- A lightweight hybrid CNN–Transformer model that delivers competitive multi-class dermoscopic classification performance on HAM10000, while surpassing EfficientNet-B3 and ViT-B/16 baselines in terms of balanced accuracy and minority class F1-scores.
- Empirical findings on whether the architecture with the highest accuracy also yields the most

clinically interpretable Grad-CAM and SHAP explanations, thereby adding new contributions to the literature on explainability for skin lesion analysis.
- A systematic explainability analysis proving the model predictions correspond to the ABCDE rule, providing evidence of clinically grounded decision-making.
- Practical design guidelines on class imbalance handling, architecture selection, and explainability method choice.
- Preliminary evidence on the feasibility of few-shot and zero-shot learning strategies for dermoscopic classification of minority disease categories.

Deliverables:

1. **Research Paper:** A paper in IEEE format that documents methodology, experimental results, and explainability analysis.
2. **Trained Models & Code Repository:** Reproducible model weights and full training, evaluation, and XAI pipelines hosted on a public GitHub repository.
3. **Explainability Visualization Report:** Grad-CAM, SHAP, and attention map visualizations with clinical alignment commentary across all seven disease categories.
4. **Web Application:** A Flask/Streamlit platform that inputs dermoscopic images and returns the predicted class, confidence score, and Grad-CAM heatmap overlay.

11 Ethics Clearance

This research will use exclusively publicly available, anonymised dermoscopic image data. The HAM10000 dataset was collected in accordance with appropriate clinical ethics protocols at the Medical University of Vienna (Protocol No. 1804/2017) and the University of Queensland (Protocol No. 2017001223) [1], and is publicly distributed through the ISIC with all patient-identifying information removed prior to release. No new patient data will be collected as part of this project. The specific protections that apply are as follows:

- **No primary data collection:** The research relies entirely on pre-existing, publicly available, de-identified datasets. No human participants will be recruited, interviewed, or examined.
- **No re-identification risk:** The HAM10000 images are distributed without patient identifiers. EXIF metadata was cleared from all images prior to release. Patient-level data splitting during preprocessing prevents information leakage without enabling re-identification.
- **Responsible model deployment:** The web application will include clear disclaimers stating that the system is a research prototype not validated for clinical use, and that all clinical decisions must remain the responsibility of a qualified healthcare professional.
- **Bias awareness and transparency:** The dataset has known demographic biases, as images were collected predominantly from patients of lighter skin tones in European and Australian clinical settings. These limitations will be explicitly acknowledged in all publications and outputs.
- **Institutional ethics compliance:** Ethics clearance will be sought from the University of KwaZulu-Natal prior to the commencement of the research.

12 Time Table

Table 2 presents the Gantt chart summarising planned project activities and their respective time frames.

Table 2: Gantt Chart overview of planned timeline

Activity	Apr	May	Jun	Jul	Aug	Sep	Oct
Literature Review							
Proposal Finalisation							
Dataset & Preprocessing							
Architecture Design							
Model Implementation							
Web App Development							
Testing & Demo Prep							
Software Demo							
Evaluation & Ablation							
Explainability Analysis							
Paper Writing							
Final Edits & Submission							
Presentation Prep							

13 Estimated Budget

Table 3 presents the estimated budget of this research project.

Table 3: Estimated project budget

Item	Justification	Estimated Cost (ZAR)
Google Colab Pro(6 months)	GPU access for model training and evaluation	R 1200.00
Kaggle GPU credits	Additional resources for ablation study	R 0.00 (free tier)
Academic journal access	Literature review and paper sourcing	R 0.00 (UKZN library)
Overleaf (12 months)	Latex Writing and supervisor collaboration	R 0.00 (institutional)
Streamlit Cloud hosting	Web application deployment	R 0.00 (free tier)
External hard drive (2 TB)	Model checkpoints and data backups	R 1000.00
Total		**R 2200.00**

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Glossary

CNN-Transformer A hybrid deep learning architecture that combines a Convolutional Neural Network (CNN) for local spatial feature extraction with a Transformer encoder for global context modelling via self-attention, exploiting the complementary strengths of both paradigms for improved image classification.

Dermoscopic Images Images captured using a dermatoscope, a non-invasive optical instrument that provides magnified, illuminated views of subsurface skin structures not visible to the naked eye, serving as the standard imaging input for automated skin lesion classification systems.

Detection of Skin Diseases The automated identification and classification of dermatological conditions from clinical or dermoscopic images using computational methods, with the goal of supporting early diagnosis and reducing reliance on specialist expertise.

Explainable Analysis The application of interpretability techniques to deep learning models to reveal how and why a prediction was made, enabling clinicians and researchers to verify that model decisions are grounded in clinically meaningful features rather than spurious correlations.

Lightweight Model A deep learning model designed to minimise computational cost in terms of parameter count, memory footprint, and inference time, enabling deployment in resource-constrained environments such as mobile devices or low-cost hardware in rural clinical settings.